

Ryubu Hosoki

Ph.D. in Science

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Research Distributed Deep Learning, GPU Programming,

Interests High Performance Computing (HPC)



Education

2023.04 - 2026.03 Ph.D. in Science

Dept. of Mathematical and Computing Science, Institute of Science Tokyo.

Thesis: A Study on Parallelization Strategies and Communication Optimizations for Accelerating Distributed Deep Learning

2021.04 - 2023.03 Master of Science

Dept. of Mathematical and Computing Science, Tokyo Institute of Technology.

2017.04 - 2021.03 Bachelor of Science

Dept. of Mathematical and Computing Science, Tokyo Institute of Technology.

Employment History

2026.04 - Present AI Laboratory, Fujitsu Research.

2024.12 - 2026.03 High Performance Big Data Research Team, RIKEN R-CCS, Student Trainee.

2024.10 - 2026.03 Institute of Integrated Research Supercomputing Research Center, Institute of Science Tokyo, Research Assistant.

2024.10 - 2025.01 Nefrock Inc., Application Development Internship.

2024.09 Computing Laboratory, Fujitsu Research, 3-week Research Internship.

2023.06 - 2023.08 Maison de la Simulation, The French Alternative Energies and Atomic Energy Commission (CEA), Research Internship.

2023.04 - 2024.09 Global Scientific Information and Computing Center, Tokyo Institute of Technology, Research Assistant.

2021.07 - 2023.03 Industrial Science and Technology, National Institute of Advanced (AIST), Research Assistant.

Scholarships

2024.04 - 2026.03 Tokyo Tech Program for Development of Next-Generation Front-Runners with Comprehensive Knowledge and Humanity (Tokyo Tech SPRING), Covering living expenses.

2023.04 - 2024.03 Cross the border! Tokyo Tech Pioneering Doctoral Research Project, covering living expenses.

2023.04 Tokyo Tech Tsubame Scholarship for Doctoral Students, covering living expenses (declined).

Teaching

2024 Computer Systems, Institute of Science Tokyo, Math. and Comp. Science, Teaching Assistant.

2023 Computer Systems, Tokyo Institute of Technology, Math. and Comp. Science, Teaching Assistant.

Publications

Conference Proceedings (Refereed)

- Ryubu Hosoki, Kento Sato, Toshio Endo, Julien Bigot, Edouard Audit, “Optimizing Intra-Layer Parallel Communication for LLM Training on Systems with Fully-Connected Mesh GPU Topology,” in *Proceedings of the Supercomputing Asia and International Conference on High Performance Computing in Asia Pacific Region (SCA/HPCAsia 2026)*, Osaka, January 2026. DOI: 10.1145/3773656.3773675
- Ryubu Hosoki, Kento Sato, Toshio Endo, Julien Bigot, Edouard Audit, “An Optimization Technique for Hiding Communication Costs of 3D Parallel Training in Deep Learning,” in *Proceedings of the IEEE 25th International Symposium on Cluster, Cloud and Internet Computing (CCGrid 2025)*, Tromsø, May 2025. DOI: 10.1109/CCGRID64434.2025.00044
- Ryubu Hosoki, Toshio Endo, Takahiro Hirofuchi, Tsutomu Ikegami, “AshPipe: Asynchronous Hybrid Pipeline Parallel for DNN Training,” in *Proceedings of the International Conference on High Performance Computing in Asia-Pacific Region (HPC Asia 2024)*, Nagoya, January 2024. DOI: 10.1145/3635035.3635045

Domestic Workshop

Refereed Papers

- Taiki Osawa, Toshio Endo, Ryubu Hosoki. cross-disciplinary Workshop on Computing Systems, Infrastructures, and Programming(xSIG 2024), Tokushima, July 2024. (Written in Japanese)

Unrefereed Papers

- IPSJ SIG HPC technical reports (2021–2025): 3 first-author papers and 3 co-authored papers.

Poster Presentations

- Ryubu Hosoki, Kento Sato, Toshio Endo, Julien Bigot, Edouard Audit, “An Optimization Pass for Training Speed-up and Strategy Search in 3D Parallelism,” in *Proceedings of the IEEE International Conference on Cluster Computing Workshops (CLUSTER Workshops 2024)*, Kobe, September 2024. DOI: 10.1109/CLUSTERWorkshops61563.2024.00031 (Refereed)

Awards and Honors

- **RIKEN OHBU Award**, RIKEN, “A CEA–RIKEN Collaborative Achievement in Optimizing 3D Parallel Deep Learning Performance”, 2026/03
- **IEEE CCGrid 2025 Best Paper Award**, IEEE Computer Society, “An Optimization Technique for Hiding Communication Costs of 3D Parallel Training in Deep Learning”, 2025/05

Skills

Languages Japanese (native), English (intermediate)

Coding L^AT_EX, C/C++, Python, CUDA, Scala, Ruby, Linux